

QUANTUM COMPUTING · BEGINNER NOTES

From Chances to Understanding

Connecting quantum probabilities with real experimental results

07 Chances and Actual Counts

When we measure a qubit in a superposition, we cannot predict the exact result of one individual measurement. What we **can** predict is how likely each result is. If we repeat the same experiment many times, the actual number of results tends to follow those predicted probabilities.

1. Probability vs. actual count

A **probability** tells us what fraction of outcomes we expect in the long run.
An **actual count** tells us how many times a particular result really appeared in our experiment.

Term	Meaning	Example
Probability	Expected chance of an outcome	50% chance of getting 0
Count	Number of times the outcome actually occurred	52 zeros out of 100 measurements
Frequency	Actual count divided by total trials	52 / 100 = 52%

2. Example: measuring $|+\rangle$

Recall that the Hadamard gate creates the state $|+\rangle$:

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

The two outcomes have equal probability:

0	50%
1	50%

3. Run the experiment 10 times

Suppose we prepare $|+\rangle$ and measure it 10 times. We might get:

0, 1, 1, 0, 1, 0, 0, 1, 1, 0

Now count the results.

Result	Actual count	Actual frequency	Expected probability
0	5	5 / 10 = 50%	50%
1	5	5 / 10 = 50%	50%

Notice: The actual counts do not have to be perfectly equal every time. They simply tend to get closer to the predicted probabilities when the number of trials becomes large.

4. What if we repeat it 100 times?

With 100 measurements, we might obtain something like:

0 → 48 times | 1 → 52 times

So the experimental frequencies are:

0	48%
1	52%

These values are very close to the theoretical prediction of **50% and 50%**.

5. More trials → better estimate

Imagine repeating the experiment:

Number of trials	Possible % of 0	Possible % of 1
10	40%	60%
100	48%	52%
1,000	49.2%	50.8%
10,000	≈ 50%	≈ 50%

The exact numbers will vary from experiment to experiment, but larger numbers of trials generally give a more stable estimate of the underlying probability.

6. This is called sampling

Each measurement is one **sample** from the probability distribution of the quantum state.

Simple idea

Quantum state → **probabilities** → **repeated measurements** → **actual counts**

The state tells us the chances. The experiment gives us the actual outcomes.

7. A useful formula

The experimental frequency of an outcome can be calculated as:

$$\text{Frequency} = \text{Number of times outcome occurs} \div \text{Total number of trials}$$

To write it as a percentage:

$$\text{Percentage} = (\text{Count} \div \text{Total trials}) \times 100$$

8. Example with an unequal probability

Not every quantum experiment gives a 50–50 result. Suppose a state has:

$$P(0) = 75\% \quad P(1) = 25\%$$

If we measure it 100 times, we might get approximately:

0 ~75

1 ~25

The actual result could be 73 zeros and 27 ones, or 77 zeros and 23 ones. The important point is that the counts should generally reflect the predicted probabilities over many trials.

9. Circuit view

$|0\rangle$

H

M

Prepare $|+\rangle$ with H, then measure

If this circuit is run repeatedly, the measurement results are collected as a set of counts.

Example: counts = {0: 498, 1: 502}

Out of 1,000 shots, this would give approximately 49.8% zeros and 50.2% ones.

Key idea

Probability describes the expected chances. **Counts** are what the experiment actually produces. More repeated measurements usually make the observed frequencies approach the predicted probabilities.

08 Check Your Understanding

Before moving forward, use these questions to check whether the main ideas about qubits, gates, superposition, measurement, probability, and counts are clear.

01 · What can a single-qubit measurement return?

A. Only 0

B. Only 1

C. 0 or 1

D. Any decimal number

Answer: C — 0 or 1.

When measured in the computational basis, a single qubit gives a classical result of either 0 or 1.

02 · What does the X gate do to $|0\rangle$?

A. Changes $|0\rangle$ to $|1\rangle$

B. Changes $|0\rangle$ to $|+\rangle$

C. Measures the qubit

D. Does nothing

Answer: A.

The X gate acts like a quantum NOT: $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$.

03 · What does H do to $|0\rangle$?

A. $|1\rangle$

B. $|+\rangle$

C. $|-\rangle$

D. Nothing

Answer: B — $|+\rangle$.

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

This gives equal measurement probabilities: $P(0) = 50\%$ and $P(1) = 50\%$.

04 · If a qubit is in $|+\rangle$, what happens when you measure it?

A. It always gives 0

B. It always gives 1

C. It gives 0 or 1 with equal probability

D. It gives 0.5

Answer: C.

The state $|+\rangle$ has equal probability for the computational-basis outcomes 0 and 1.

05 · Does 50% probability mean the next measurement must be 0?

A. Yes

B. No

Answer: B — No.

A probability describes a chance, not a guaranteed next result. A 50% probability means that over many repeated measurements, the results tend to approach a 50–50 distribution.

06 · A circuit is measured 100 times and produces 47 zeros and 53 ones. Is this reasonable for a 50–50 state?

A. Yes

B. No

Answer: A — Yes.

The theoretical prediction is 50 zeros and 50 ones on average, but real finite experiments fluctuate. 47/100 and 53/100 are both reasonably close to 50%.

07 · Which statement correctly describes probability and count?

A. Probability is the actual number of results

B. Count is the predicted chance

C. Probability describes chances; count records outcomes

D. They are always exactly the same

Answer: C.

Probability is the theoretical expectation, while count tells us what actually happened in a particular set of measurements.

08 · What happens to the estimate of probability as we perform more trials?

A. It usually becomes more stable

B. It becomes completely random

C. It always becomes exactly 50%

D. It disappears

Answer: A.

With more trials, the observed frequency generally gives a better estimate of the underlying probability.

09 · If $P(0) = 75\%$ and $P(1) = 25\%$, what would you expect from about 1,000 measurements?

A. About 500 zeros and 500 ones

B. About 750 zeros and 250 ones

C. About 250 zeros and 750 ones

D. Only zeros

Answer: B.

Since the probability of 0 is 75%, we expect roughly 750 zeros. Since the probability of 1 is 25%, we expect roughly 250 ones.

10 · What does measurement do to a superposition?

A. It returns a classical outcome such as 0 or 1

B. It always returns both 0 and 1

C. It creates another qubit

D. It removes probability from the universe

Answer: A.

Measurement converts the quantum information into a classical measurement outcome. For computational-basis measurement of one qubit, that outcome is 0 or 1.

Mini Challenge

Try solving this before looking at the answer.

You prepare a qubit in $|0\rangle$ and apply H:

$|0\rangle$

H

M

You run the circuit **200 times**.

You observe:

0 → 103 times | 1 → 97 times

Question: Is this result consistent with the expected behavior of $H|0\rangle$?

Yes.

$H|0\rangle$ creates $|+\rangle$:

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Therefore:

P(0) = 50% P(1) = 50%

Out of 200 measurements, we would expect roughly 100 zeros and 100 ones.

The observed values of 103 and 97 are close to that expectation, so the result is reasonable.

What you should know now

Concept	Remember
Qubit	Can exist in a quantum state involving $ 0\rangle$ and $ 1\rangle$
X gate	Flips $ 0\rangle \leftrightarrow 1\rangle$
H gate	Creates equal superposition from $ 0\rangle$ or $ 1\rangle$
Measurement	Produces a classical outcome such as 0 or 1
Probability	Describes how likely each outcome is
Count	Records how many times an outcome actually occurred
More trials	Usually give a better estimate of the predicted probabilities

Final takeaway

Quantum computing does not tell us exactly which result will occur in every measurement. Instead, the quantum state determines a **probability distribution**. When we repeat the experiment many times, the **actual counts** reveal that distribution experimentally.